Analyzing the Effect of Network Width on Graph Transformers

All experiments so far, and the reproductions of Sanford et al. 2024a and Yehudai et al. 2025

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The question and the set-up

How does the width a graph transformer needs to *learn* connectivity grow with graph size n?

- **Task:** connectivity matrix: for every pair of nodes, are they in the same component?
- **Model:** encoder, one token per node (its row of $A + I$), pre-LN, pairwise read-out $HWH^\top$
- **Width** m = embedding dimension; heads of size 8 ($m/8$ heads), feed-forward $4m$
- **Critical width m^*:** smallest m whose final test pair accuracy reaches 0.95, interpolated in $\log_2 m$ per seed, then averaged
- **Final protocol:** 128 000 graphs, 24 000 steps, batch 128, LR 10^{-3}, depth 6, 2 seeds; LR chosen on validation, numbers reported on test

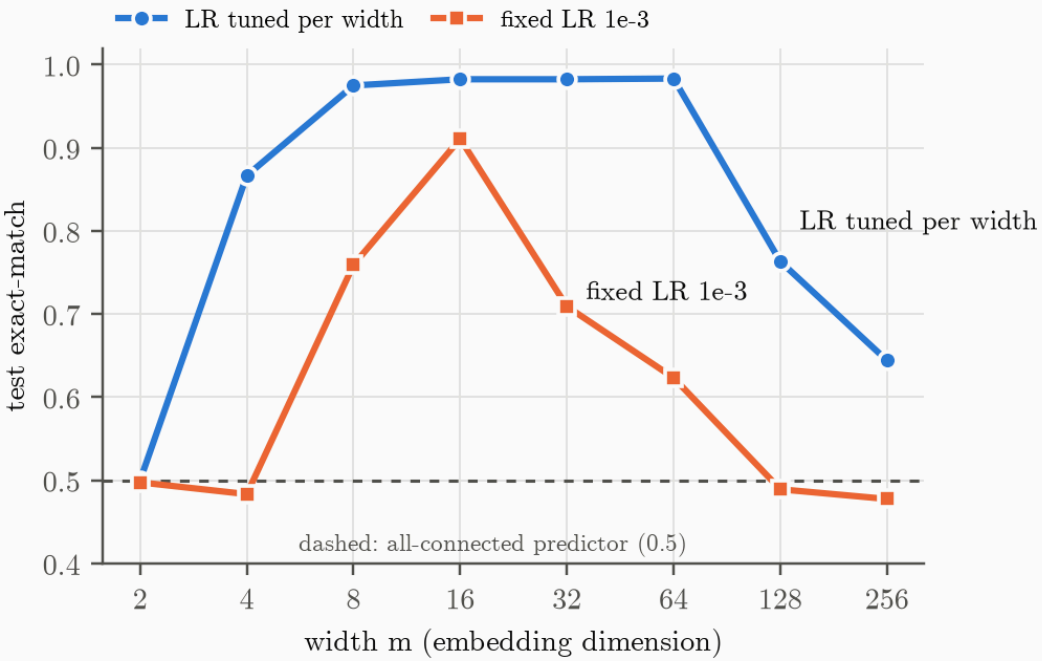
Theory: width $\approx \sqrt{N}$ suffices to **represent** connectivity at depth $\log N$ (Sanford et al. 2024a); width $O(n)$ at constant depth for 1-vs-2 cycles (Yehudai et al. 2025). We measure the width to **learn** it.

Headline results

result	what it says
$m^* \propto n^{1.74}$	critical width at depth 6, n = 32–56 (90 %: 1.59–1.90)
$m^* \propto n^3$ at depth 2	training does not find the constant-depth, linear-width solution
$n^{1.27}$ / $n^{1.03}$	growth when each token gets its 2-hop / 4-hop neighbourhood
2.2–2.5×	extra width Sanford et al.'s edge tokens need (with 4× the steps)
0.995	2-hop classifier on Sanford et al.'s GraphQA data (their transformer: 0.98)
0.77	one number (edge count) on Yehudai et al.'s connectivity data
gradual, pairwise	probes: "connected" is built up in every layer, not stored per node

Part 1 — Measuring width honestly

A fixed learning rate fakes a width effect



- One fixed rate: wide models look up to **0.4 worse**
- Tuned per width: widths 8–64 all ≈ 0.97–0.98
- The best rate **falls as width grows** ($10^{-2} \rightarrow 10^{-3}$ at $m = 256$)
- The remaining drop at $m \geq 128$ is memorization: 8 000 graphs remove it (Q1c)

$n = 24$, depth 2, 2000 graphs, 3 seeds (Q1, Q1b, Q1c)

The first benchmark leaked; swap does not

shortcut in the old data	evidence
node indices reveal the components	blobs always nodes $[0, n_a)$ and $[n_a, n)$
graph statistics reveal the label	degree + closed walks $\operatorname{tr} A^3 \ldots \operatorname{tr} A^6$: 0.68–0.90
small graphs are local	a 6-hop check is exact on 60–100 % of graphs

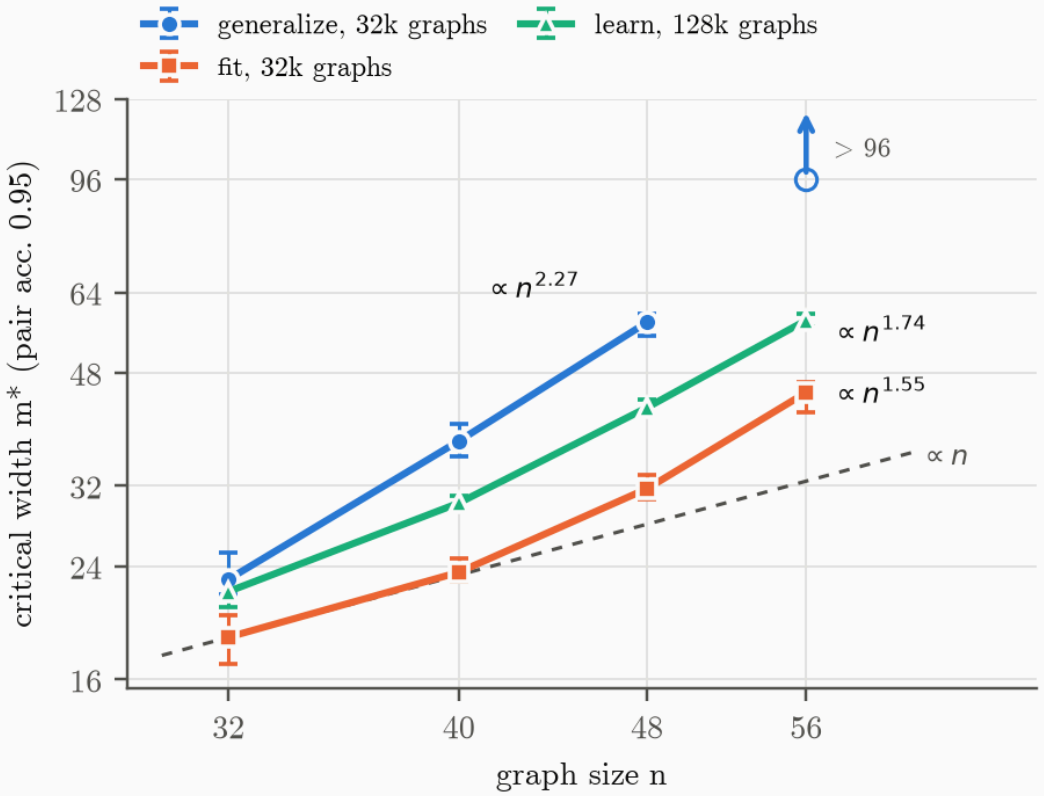
swap: two sparse blobs; pick a_1, a_2 and $b_1, b_2 \geq 6$ hops apart. Disconnected: add $a_1 a_2$, $b_1 b_2$. Connected: add $a_1 b_1$, $a_2 b_2$. Same degrees, every new cycle ≥ 7 edges, labels shuffled.

n	32	40	48	56	64	128
statistics $\longrightarrow$ label	0.53	0.51	0.49	0.50	0.51	0.51
4-hop check, exact	0	0	0	0	0	0
diameter (connected)	8.6	9.8	10.9	11.7	12.5	16.7

Trivial predictor: 0.5. On the leaky data width 8 sufficed at $n = 24$.

Part 2 — How width must grow with n

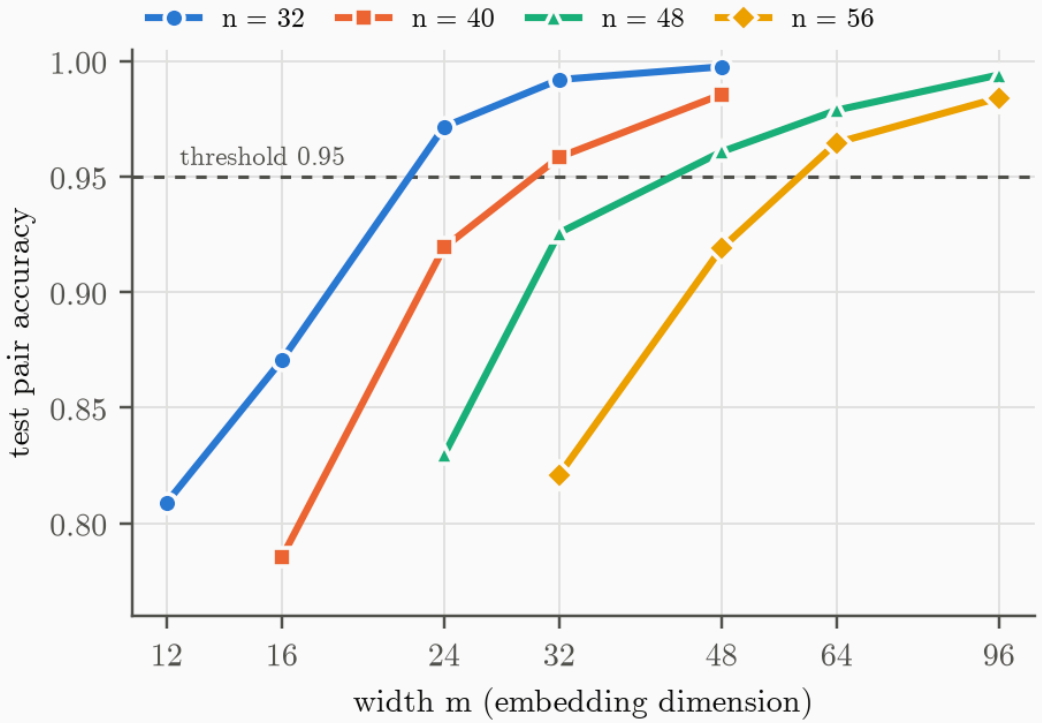
Enough data closes the fit–generalize gap



- 32 000 graphs: fit $\propto n^{1.55}$, generalize $\propto n^{2.27}$; the gap widens with n
- 128 000 graphs at the same 24 000 steps: the two coincide (within ≈ 2)
- The gap was a data ceiling, as in Q1c
- Below the threshold nothing learns: $n = 64$ needs $m \geq 64$; at $n = 128$ nothing up to 128 leaves the trivial predictor

depth 6, LR $3 \cdot 10^{-3}$, 2–3 seeds

Main result: $m^* \propto n^{1.74}$



n	fit	generalize	m^*/n
32	20.9	21.9	0.68
40	28.2	30.1	0.75
48	38.5	42.4	0.88
56	52.7	57.9	1.03
growth	$n^{1.65}$	$n^{1.74}$	

90 % bootstrap range 1.59–1.90 (one seed per n; only 16 combinations).
128 000 graphs, 24 000 steps, LR 10^{-3}, depth 6.

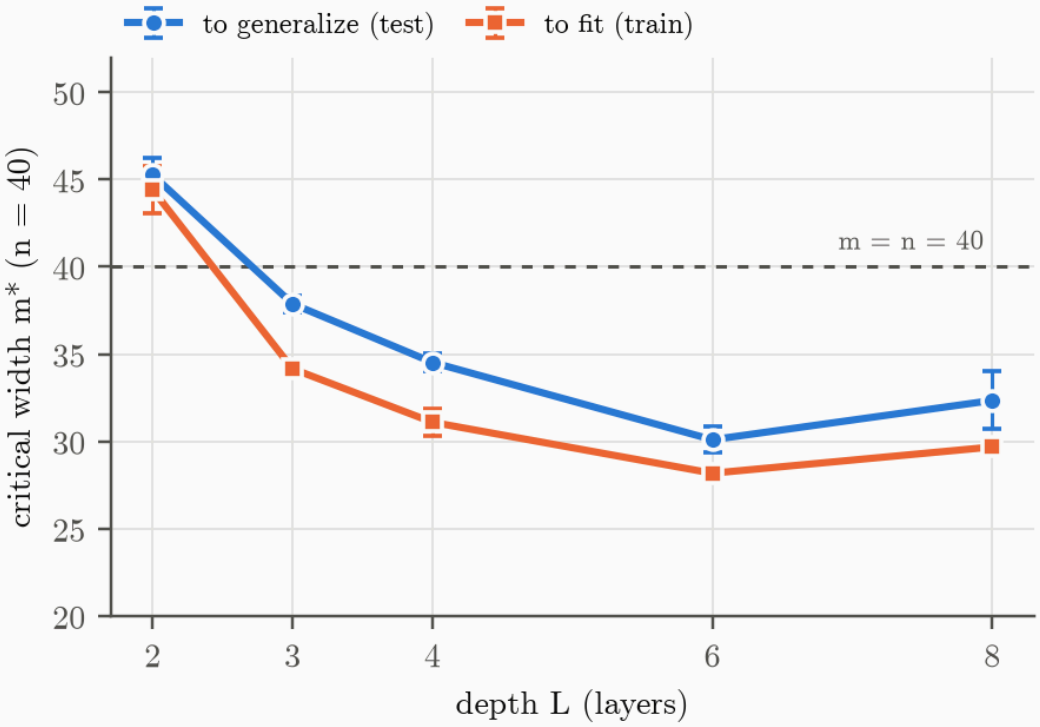
The growth survives every check

could it be…	test	outcome
too little data?	32k ⟶ 64k ⟶ 128k graphs	gap closes, **growth stays superlinear**
too little training?	2× the steps (48 000)	**critical width unchanged** (within seed spread)
the learning rate?	$3 \cdot 10^{-3}$ vs 10^{-3}, tuned on validation	helps wide models at $n = 56$; slope 1.94 ⟶ **1.74**
the n-wide input?	fixed 96-wide random projection	**still doubles** from $n = 40$ to 56 ($n^{2.03}$)
noise?	per-seed crossings	**seeds within ≈ ± 3**

Over $n = 32$–56, the width needed to **learn** connectivity grows faster than n, while the width needed to **represent** it is about $\sqrt{N}$.

Part 3 — Can depth replace width?

At $n = 40$, depth saves about a third

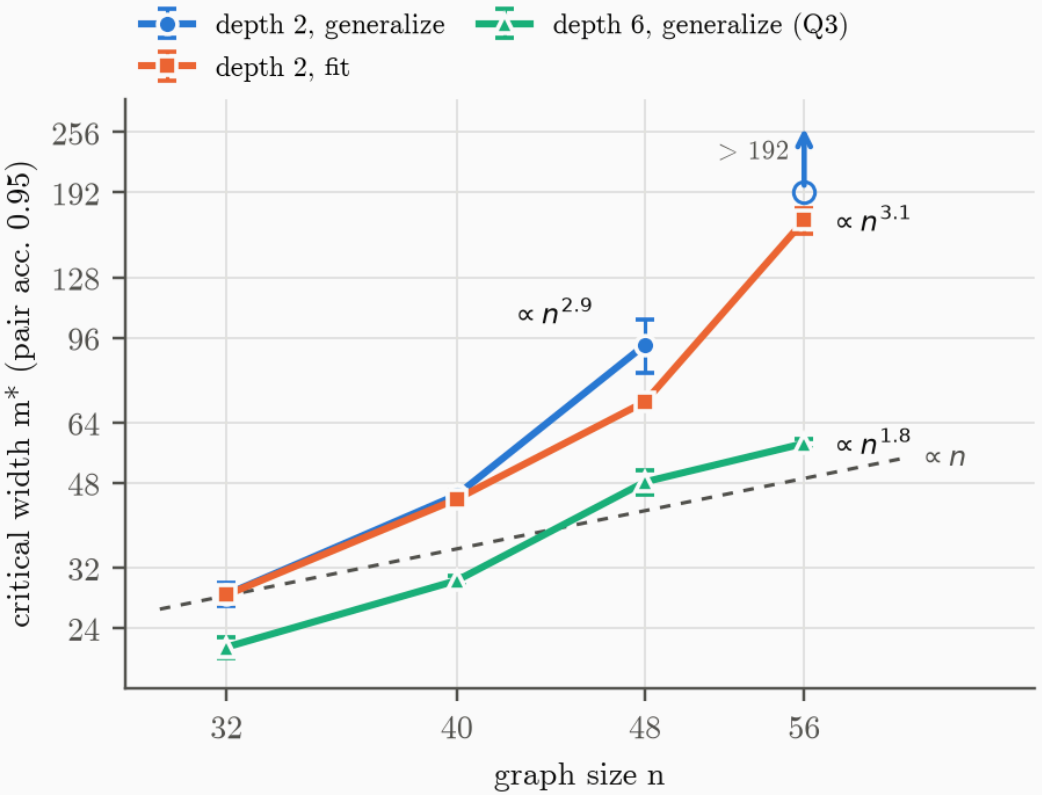


L	2	3	4	6	8
m^*	45.3	37.9	34.5	30.1	32.3

- **2 layers learn** at $m \approx 1.1n$ on graphs of diameter ≈ 10: the model does not follow paths hop by hop
- Beyond 4–6 layers depth buys nothing
- $L = 6$ repeats the main result exactly (30.1)

$n = 40$, 128 000 graphs, LR 10^{-3}, 60 runs (Q7a)

At depth 2 the critical width grows as n^3



	32	40	48	56
$L = 2$, fit	28.1	44.4	70.7	168
$L = 6$	21.9	30.1	42.4	57.9
ratio	1.3	1.5	1.8	> 3

- Fit $\propto n^{3.07}$: a linear-width, constant-depth solution exists, but training does not find it as n grows
- At $n = 56$, two layers do not generalize to 0.95 by width 192
- "Depth trades mildly" holds only at $n = 40$

Q7c, 34 runs

Part 4 — Reproducing Yehudai et al. 2025

What Yehudai et al. prove

result	statement
Thm 4.1	1 cycle vs 2 cycles: 2 layers, width $O(n)$, adjacency-row tokens
Thm 4.2	lower bound: $mpHL = \Omega(n)$ for the same task, unconditional
Thm 4.3	A^L with $O(L)$ layers: one hop per layer (not repeated squaring)
general connectivity	only a sketch: width $O(n \log^3 n)$ via linear sketches (Ahn, Guha, McGregor 2012)
experiments	**fitting** criterion, 5 000 graphs, depth 1, fixed LR grid

Their theory is about **representing**; their experiments measure **fitting**, on data we show is easy.
Our results measure **learning** on audited data.

Reproduction: their connectivity data is easy

Their code (two bugs patched: a `NameError`, and the cache reloading the **train** split as validation and test)

$n = 50$, 1 layer, 100 graphs	test
adjacency rows	**1.00**
edge list	0.93
our model on **their** data	**1.00** by epoch 4

Why it is easy

- Classes come from different generators (ER, geometric, scale-free, SBM)
- Edge count **alone** classifies **77 %** mean degree too
- Fixed $n = 50$: column j always means the same node

On degree-matched data the same model stalled at 0.59–0.70; on `swap` it needs the widths of Part 2.

Part 5 — Reproducing Sanford et al. 2024a

Their claim and their numbers reproduce

- **Claim:** connectivity at depth $O(\log N)$ with width $\approx \sqrt{N}$ (vertex + edge + task tokens, $N = n + |E|$); one layer needs $mH = \tilde{\Omega}(N)$
- **Experiment:** GraphQA connectivity (ER graphs, $n \leq 19$, one s–t question), decoder $L = 12$, $m = 768$, trained on 1K and 100K graphs. No code released: we re-implemented data, tokens and model

	theirs	**ours**
test accuracy, 1K graphs	92.9	**93.0**
test accuracy, 100K graphs	98.0	**98.2** (LR 10^{-4})
their LR $5 \cdot 10^{-4}$ at 100K	—	**stuck at 0.93, then NaN**

But GraphQA connectivity is local

	test	hard: ≥ 4 hops	hard: no path
degree rule	0.973	1.00	0.00
3-hop rule	0.981	1.00	0.00
trees: size	0.944	0.878	0.720
1. degrees	0.977	0.938	0.411
1. 2-hop	**0.995**	0.932	0.893

Gradient-boosted trees on 11 features within 2 hops of s and t, 50k training examples.

- A 2-hop classifier **beats the 71M-parameter transformer** (0.982)
- 94 % of connected pairs are ≤ 2 hops apart
- **Width only buys the shortcut** (B1, m = 8–768): 0.98 from $m = 32$, flat to 768; on the hard set every width sits below the 2-hop classifier

Their encoding on audited data does not learn

- **B2:** `swap`, n = 32–56, depth 6, 128 000 graphs, 24 000 steps, their tokens with 32 questions per graph: **no width up to 96 leaves "always yes"**
- **b2long:** 100 000 steps, m up to 192: the loss escapes, but **only into memorization** (test below "always yes")
- **Confound:** 32 labels per graph vs n^2 = 1024–3136 in our task

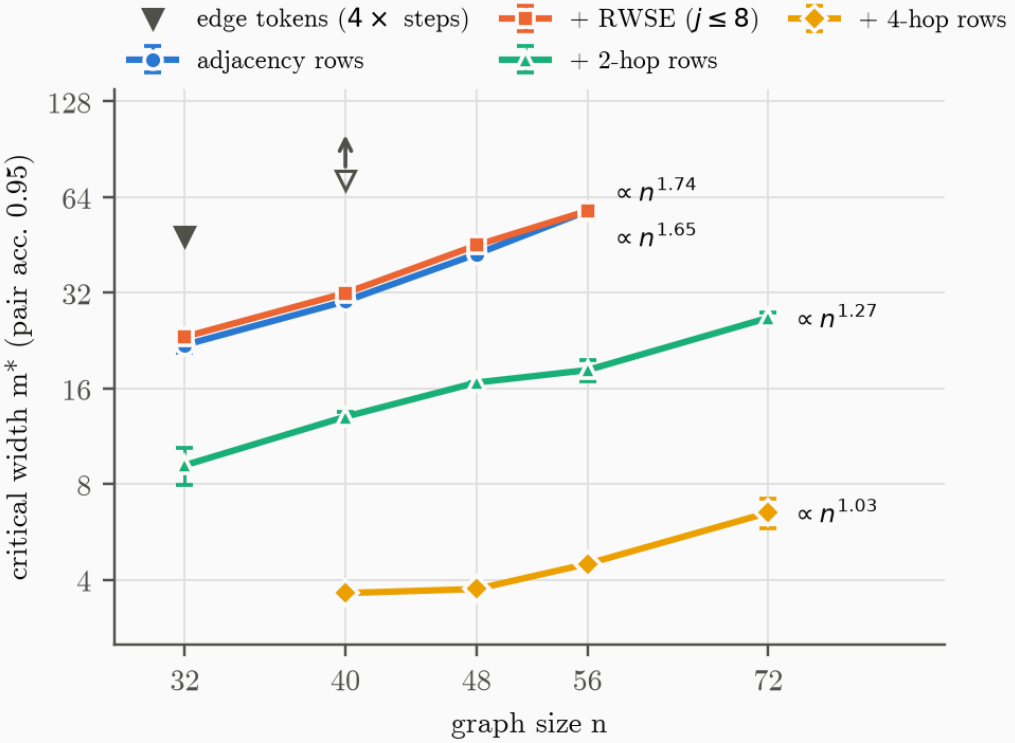
Fix — edge tokens with all-pairs labels (same read-out as our task):

	$n = 32$	$n = 40$
adjacency rows, 24 000 steps	21.9	30.1
edge tokens, 24 000 steps	no $m^* \le 96$	no $m^* \le 96$
edge tokens, 96 000 steps	**48.2** (2.2×)	**≥ 72** (≥ 2.4×)

Even with 4× the steps, ≥ 5× their $\sqrt{N}$ for this very encoding.

Part 6 — Input encodings

The hops left to compute set the width

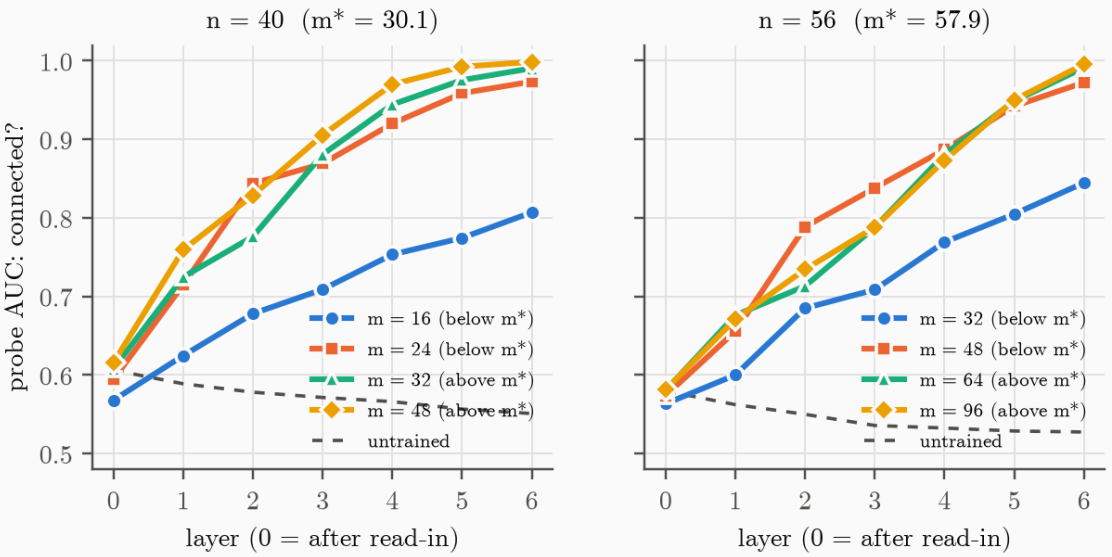


input	$n = 56$	growth
adjacency rows	57.9	$n^{1.74}$
1. RWSE ($j \le 8$)	57.9	$n^{1.65}$
1. 2-hop rows	18.3	$n^{1.27}$
1. 4-hop rows	4.5	$n^{1.03}$

- 2 free hops cut the width 2.3–3.2×
- RWSE adds nothing; its usual 16-step version leaks connectivity (0.90 at $n = 40$)
- `swap` diameter grows with n, so part of the $n^{1.74}$ is path length

Part 7 — What does each layer compute?

Probing: “connected” is built up gradually



- Pair-probe AUC for “connected” rises in **every** layer; ≥ 0.95 only at layer 4–6
- **No hop hierarchy:** 3-hop info appears at layer 1 and stays flat; 1-hop fades
- **Not stored per node:** one node’s vector gives its reachable set at AUC ≤ 0.89; a pair of vectors gives 0.99–1.00
- Models far below m^* fall behind after 1–2 layers

16 saved models, n = 40 / 56, linear probes on the residual stream; dashed: untrained

What it means

Takeaways

1. **Measuring width is a protocol problem first.** LR, data and steps each interact with width enough to fake or hide an effect.
2. **Benchmarks decide the answer.** Both reproduced papers use data a local rule solves (Yehudai: edge count 77 %; Sanford: 2-hop 0.995). An audit is a contribution in itself.
3. **Representation ≠ learning.** The width to learn connectivity grows as $n^{1.74}$ at depth 6 and $\approx n^3$ at depth 2, far above the representation bounds.
4. **Path length is a large part of it.** Free hops lower the exponent: $1.74 \longrightarrow 1.27 \longrightarrow 1.03$.
5. **Mechanism:** a component signature compared pairwise, refined in every layer; no path doubling.

Limitations

- Sizes span less than 2× (n = 32–56), 2 seeds; local slopes vary (≈ 1.2–2.9)
- The bootstrap has only 16 seed combinations
- One task (connectivity), one architecture family
- Linear probes show what is easily readable, not what the model uses

Thesis status and questions

Written (Mongolian, department template)

- Introduction, Ch. 1 theory, Ch. 2 method, Ch. 3 scaling

To write

- Ch. 4 depth (Q7a, Q7c)
- Ch. 5 encodings + both reproductions
- Probing results, conclusion

Questions

1. Ch. 4 (depth) as its own short chapter, or part of Ch. 3?
2. Where should the reproductions go: Ch. 5, or a separate chapter?
3. Is the $n = 32$–56 range enough, or add one larger size?
4. Planned Q2 / Q4 (task hierarchy, Graphormer bias): in scope?